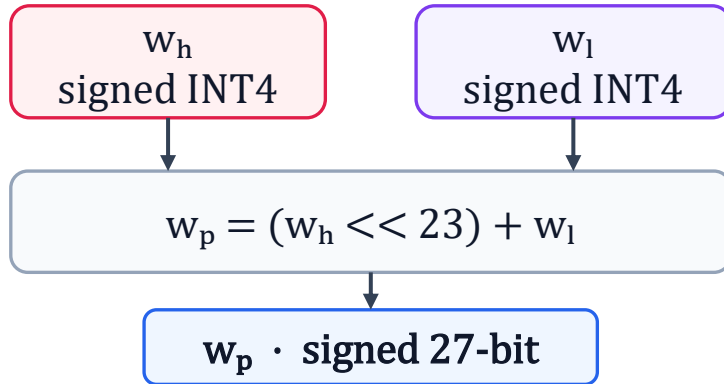


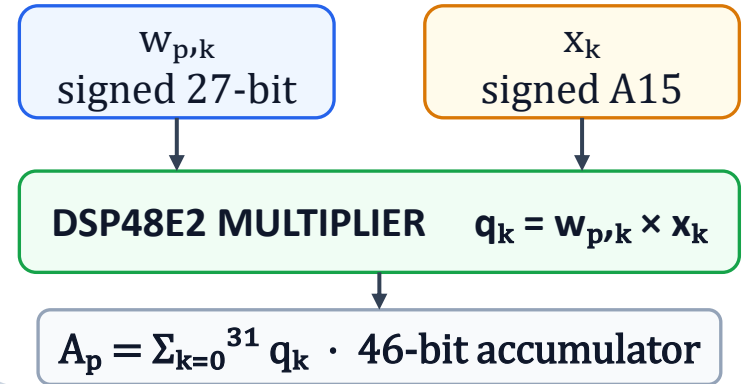
TWO-LANE INT4 PACKED MULTIPLICATION ON DSP48E2

Two signed INT4 products share one DSP; 23-bit lane spacing prevents product overlap

A Signed weight packing



B Packed multiply and G32 accumulation



C Lane extraction and sign correction

